AetherGuard

AI-powered sensor integrity and landslide early warning for the Himalaya and the Western Ghats

Technical report

What was built, on which data, how every model works, how accurate it is, why traditional systems fail, how AetherGuard addresses those failures, and how the system can be improved.

Status. Research prototype. Built end to end on real, open data and run on a single Windows laptop. **Not validated for issuing public warnings.**

Region. South Asia: the Himalaya (India, Nepal, Bhutan) and the Western Ghats (India). 2007-2024, daily, 0.05° (about 5 km) cells.

Evidence base. Every figure in this document comes from the project's own result files (reports/*.json and *.md) produced by the pipeline; the main source is reports/FINAL_REPORT.md. Where a result is a shortfall, it is reported as one.

Date. 5 October 2026

Contents of this document: 1 Executive summary · 2 Why traditional systems fail · 3 System architecture · 4 Datasets · 5 The models · 6 Results and accuracy · 7 How AetherGuard addresses the failures · 8 Limitations · 9 How to improve · 10 Reproducing and running it · Appendices (glossary, project history, file index).

Table of contents

1 Executive summary . . . 3
1.1 Headline results . . . 3

2 Why traditional systems fail . . . 5
2.1 How landslide early warning usually works . . . 5
2.2 The motivating case: Wayanad, 30 July 2024 . . . 5
2.3 The failure modes, with the evidence found in this project . . . 6

3 System architecture . . . 7
3.1 Overview . . . 7
3.2 Why a two-stage gated design . . . 7
3.3 Components . . . 8
3.4 Design principles that shaped the build . . . 8
3.5 Software structure and compute . . . 8

4 Datasets . . . 9
4.1 What was used . . . 9
4.2 Landslide labels: how the positives were built . . . 9
4.3 Frozen splits . . . 10
4.4 Grid, domain and the day basis . . . 10
4.5 Negatives and the 'universe' sample . . . 11
4.6 The feature store . . . 11
4.7 Data-quality findings that shaped the design . . . 11

5 The models . . . 12
5.0 Which model was built on which data . . . 12
5.1 Stage 1: the sensor-integrity engine . . . 12
5.2 The extreme-rain corrector . . . 13
5.3 Stage 2: the landslide hazard model . . . 13
5.4 The alert engine . . . 14
5.5 Serving and interfaces . . . 14

6 Results and accuracy . . . 15
6.1 How to read the numbers . . . 15
6.2 Stage 1: how well sensor faults are caught . . . 15
6.3 The extreme-rain corrector . . . 16
6.4 The landslide hazard model . . . 17
6.5 The alert system and the case studies . . . 18
6.6 Scorecard against the original goals . . . 19

7 How AetherGuard addresses the failures . . . 20

8 Limitations and honest shortfalls . . . 21

9 How the system can be improved . . . 21

10 Reproducing and running the system . . . 23

Appendix A Glossary . . . 24

Appendix B How the project was built . . . 24

Appendix C Key files . . . 25

1 Executive summary

AetherGuard is a two-stage early-warning pipeline for rain-triggered landslides. It was designed after studying where existing systems fail, and it was built and tested end to end with real data: gauge records from NOAA, India's IMD rainfall grid, CHIRPS satellite rain, NASA soil-wetness, Copernicus terrain, and NASA/NSIDC landslide catalogs. It runs on one laptop, ships with 1,323 automated tests, a one-command pipeline runner, a REST API and a six-page dashboard.

The idea in one paragraph. Landslide warnings depend on rainfall measurements, and existing systems fail in two linked ways. First, the rain inputs can badly under-measure extreme local storms, and nobody checks whether a rain gauge is telling the truth (a blocked gauge reads zero in a storm). Second, fixed rainfall thresholds raise many false alarms. AetherGuard therefore asks two questions in order. **Stage 1:** can this sensor reading be trusted? **Stage 2:** given trusted rain, soil wetness and terrain, how likely is a landslide on this 5-km cell today? An **alert engine** then turns the probability into Normal / Watch / Warning / Emergency, attaches a data-confidence flag and reasons, and refuses to stay silent when it is blind.

1.1 Headline results

Question	Result	Where
How many landslide days did the hazard model catch in the unseen year 2018? (target: at least 90%)	**85.0%** of 80 (95% interval 77.5-92.5). Target **not reached**; the interval still contains 90%.	§6.4
Same, for the complete alert system (model + uncertainty rule + blind-sensor rule + hysteresis)	**91.2%** (73 of 80) at the Watch level, with 27% of all cell-days on Watch or higher.	§6.5
False alarms compared with standard methods at similar detection	Model alerts on **21.4%** of cell-days; intensity-duration rule 32.4%, LHASA-style rule 39.8%, logistic regression 40.9%. The 11-point gap to the best baseline is statistically clear.	§6.4
Cross-validated over 2007-2017 (all choices made out-of-fold)	To catch 90% of events the model alerts on **18.8%** of cell-days; baselines need 32.1%, 36.1% and 37.7%.	§6.4
Kerala floods 2018 (23 labelled landslide cell-days)	**23 of 23** detected; median first alert 7 days ahead (capped by the data window).	§6.5
Wayanad 2024 disaster day (no label; never tuned on)	**Levels saturate:** the focus cell is at Emergency on all 22 days, so the system cannot single out 30 July. Reported as a limitation.	§6.5
Sensor-fault detection on unseen test stations (injected faults)	Macro recall **0.70** versus 0.30 for the best existing method, but at 4.9% false flags (0.9% without two out-of-distribution Nepal stations).	§6.2
Extreme-rain corrector	Beats raw CHIRPS at all quantiles (pinball loss -43% at the 90% quantile) but **does not close** the Wayanad gap (predicted 71 mm vs 179 mm measured by IMD).	§6.3

What works

- The model ranks risk far better than conventional rainfall rules: roughly 11 points fewer false alarms at the same detection rate, confirmed out-of-sample and in cross-validation.
- Zero-reading, offset and spike faults in rain gauges are caught several times more often than by NOAA-style quality control.
- The evaluation is honest by construction: frozen checksummed splits, a single look at the test year, bootstrap intervals, and every shortfall reported.
- The whole pipeline is reproducible with one command; every number in the final report is read from a result file.

What does not work yet

- The 90%-detection target is missed by the model alone (85%); the 91% figure needs the extra uncertainty rule.
- The alert levels cannot pick out the Wayanad disaster day: the product-level signal for such extremes is weak.
- The sensor checker over-flags stations that look unlike its training stations (the two Nepal gauges).
- Probabilities are uncertain by about a factor of two, and the test year holds only 80 positive cell-days, so most intervals are wide.

A note on 'accuracy'. Landslides are extremely rare: about 6 per million cell-days. A system that never raised any alert would be 99.9994% 'accurate' and completely useless. So this report measures **detection rate** (recall: the share of real landslide days that were flagged) against the **false-alarm share** (the share of all cell-days put on alert). Plain accuracy is not reported because it would be misleading.

2 Why traditional systems fail

2.1 How landslide early warning usually works

Most operational and research systems use one of three approaches, all of which stand on a rainfall input: (1) **rainfall thresholds** (intensity-duration curves, antecedent rain) that raise an alert when rain exceeds a fixed line; (2) **susceptibility × rainfall** rules such as NASA's LHASA, which combine a static terrain-susceptibility map with a rain trigger; and (3) **regional forecast products** built on global rain products (satellite estimates and reanalysis). India's Geological Survey launched an experimental regional landslide forecast on 19 July 2024.

All three inherit the quality of the rain measurement. None of them asks whether the rain gauge that feeds them is healthy, and the first two use fixed rules that cannot adapt to local wetness, terrain or measurement quality.

2.2 The motivating case: Wayanad, 30 July 2024

On 29-30 July 2024 landslides at Mundakkai and Chooralmala in Wayanad, Kerala killed hundreds of people. Global products measured the 48-hour rainfall at roughly ERA5 ≈ 107 mm, ECMWF IFS ≈ 106 mm and NASA MERRA-2 ≈ 80 mm, against about 572 mm reported by local gauges (a secondary source). The regional forecast had rated the area 'low' until the day, and the red alert was issued after the slides. A grid cell of about 25 km versus a point gauge explains part of such a gap, but unlikely all of a five- to seven-fold difference.

This project measured the same day again with its own data. For the IMD day of 30 July 2024 the products disagreed by more than a factor of 100, from 0 mm (CHIRPS on the UTC day) to 179 mm (IMD grid):

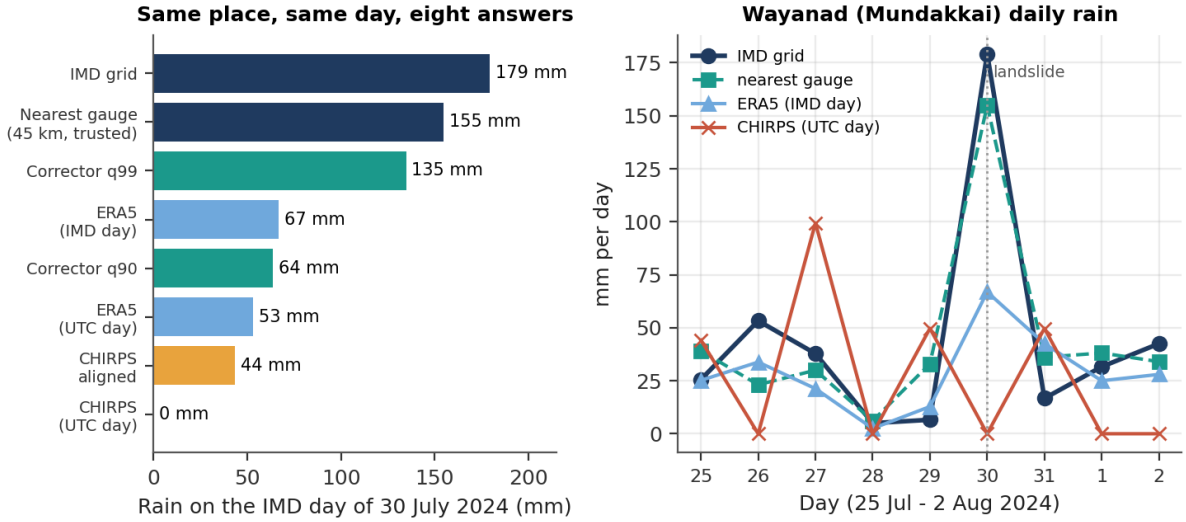


Figure 1. Eight different answers for the same place and day (left) and the daily series around the event (right). Sources: reports/phase3e_case_inputs.json. The nearest trusted gauge is 45 km from the landslide point.

2.3 The failure modes, with the evidence found in this project

#	Failure of traditional systems	Evidence (literature / design phase)	Measured in this project
F1	Global rain products under-measure extreme local (orographic) rain.	Wayanad: ERA5 / IFS ≈ 106 mm and MERRA-2 ≈ 80 mm vs ≈ 572 mm from gauges. NASA LHASA detection reportedly 8-60% depending on rain product.	30 Jul 2024, IMD day: IMD 179 mm, trusted gauge 155 mm, ERA5 67 mm (IMD day) / 53 mm (UTC day), CHIRPS 44 mm realigned / 0 mm UTC.
F2	Sensor faults go undetected or are confused with real extremes (a blocked gauge reads zero in a storm; undercatch 2-30%).	Tipping-bucket gauges clog; power and transmission gaps; GHCN has few flagged errors.	NOAA-style QA catches only 3-19% of zero-reading faults, ≈0% of drift and offset faults (TEST). Only 10 real NOAA flags exist in 2007-2024.
F3	Static thresholds give low detection and many false alarms.	Low thresholds cause many false alarms; no uniform validation criteria (Guzzetti et al., 2020).	Best rainfall-threshold baseline alerts on 32.4% of cell-days at 87.5% recall; AetherGuard 21.4% at 85.0% (unseen 2018).
F4	Inflated accuracy from random splits and biased, incomplete labels.	Landslide inventories are partial; random splits leak neighbouring days.	Labels per year vary more than 10-fold (8 in 2007 vs 127 in 2015); post-2018 labels almost absent; first model's test result was worse than its validation result (§Appendix B).
F5	No uncertainty, no degraded mode, silent failure.	Operational systems return 'normal' when inputs are missing or wrong.	Data-confidence flag; blind-sensor escalation; out-of-distribution station guard (§5.4).
+	Rain 'days' are defined differently across products.	Not widely documented.	CHIRPS/ERA5 use UTC days, IMD uses 08:30 IST. Station-vs-CHIRPS correlation peaks at lag -1 day; realigning raises it to lag 0. Kerala 2018 IMD-day totals are about double the UTC-day totals.
+	Satellite product artifacts at the event itself.	—	Raw CHIRPS files show alternating exact zeros and repeated values at Wayanad (a disaggregation artifact of the product); CHIRPS reads 0 mm on the disaster day.
+	Few labelled gauges in the mountains.	GHCN-Daily has 138 Indian and only 2 Nepali stations with data in 2015-2024.	Only 26.5% of station-days (1990-2024) have a value; most gauges report monsoon season only.

3 System architecture

3.1 Overview

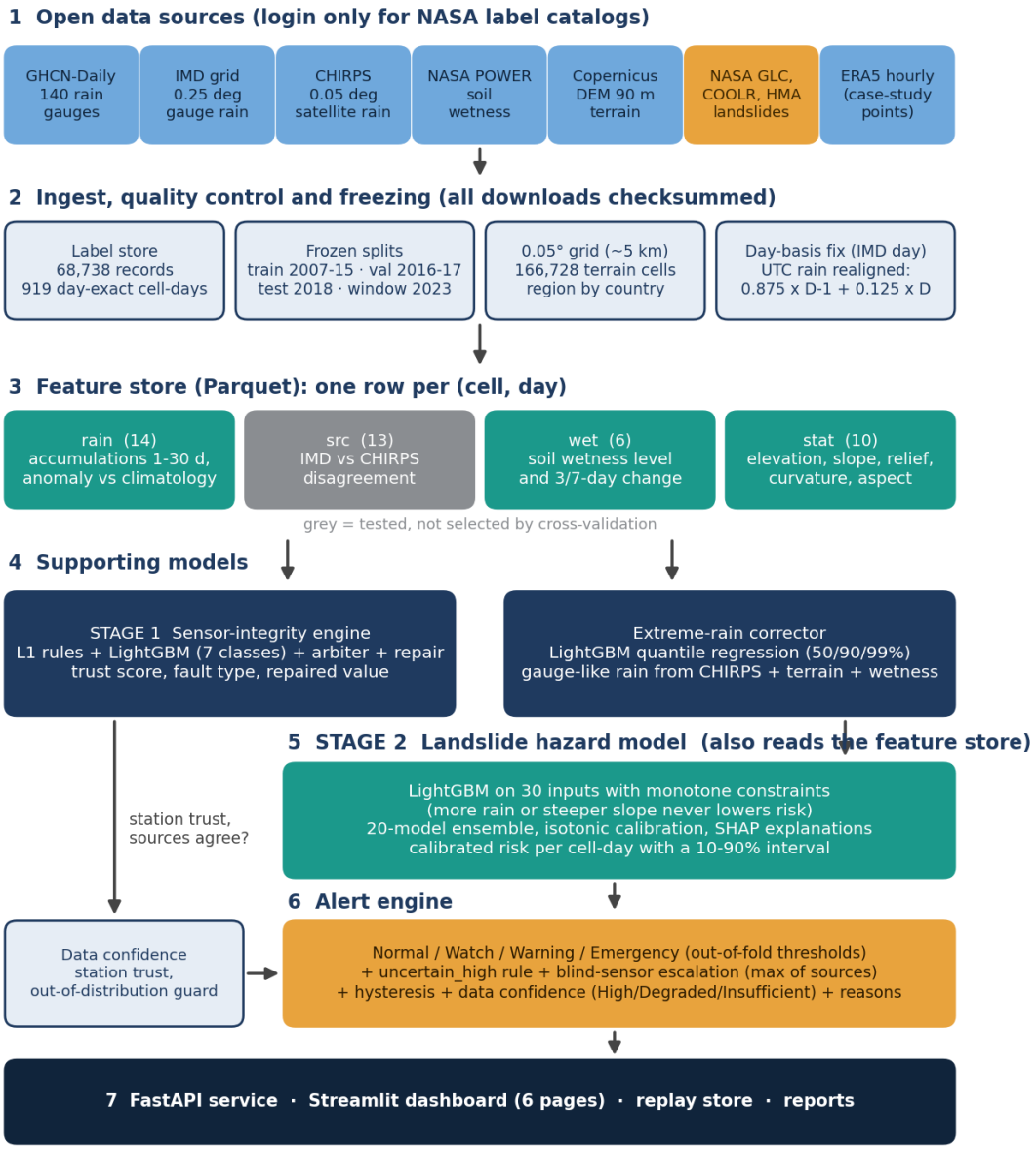


Figure 2. The AetherGuard pipeline. Grey = feature group tested but not selected by cross-validation.

3.2 Why a two-stage gated design

The alternatives were (a) a single joint model and (b) rules or physics only. A joint model would need far more labelled fault-plus-landslide data than exists and could not be audited stage by stage; rules-only is the approach that failed at Wayanad (it is kept as a baseline). Splitting the problem gives each stage one purpose, a defined interface and an independent test. Stage 1 can be validated on gauge faults without any landslide labels; Stage 2 can be validated on landslides; the alert engine joins them with explicit, auditable rules.

3.3 Components

Component	Purpose	Technique	Output
Ingest and freeze	Download, check and fix every dataset; freeze splits and samples	Checksummed manifest, exact-size checks, resumable downloads, frozen read-only Parquet	Label store, splits, samples, feature store
Stage 1: integrity engine	Judge each gauge reading	L1 deterministic rules + LightGBM (7 classes) + isotonic calibration + arbiter + repair	Trust, fault type, genuine-extreme flag, repaired value ± band
Extreme-rain corrector	Estimate gauge-like rain, especially extremes, from CHIRPS	LightGBM quantile regression (50/90/99%) + exceedance classifier	q50, q90, q99, P(exceed)
Stage 2: hazard model	Landslide probability per cell-day	LightGBM, monotone constraints, 20-member ensemble, isotonic calibration, SHAP	Risk, 10-90% interval, top-5 reasons
Alert engine	Turn risk into action levels safely	Out-of-fold thresholds, uncertain_high rule, blind-sensor rule, hysteresis, confidence states	Level, confidence, reasons
Serving	Make results usable	FastAPI (127.0.0.1), Streamlit dashboard (6 pages), replay store	REST endpoints, risk map, drill-downs
Pipeline and report	Reproduce everything	24-stage runner with reuse of verified frozen stages; final-report generator reading every number from files	reports/FINAL_REPORT.md/.json

3.4 Design principles that shaped the build

- **Recall first.** A missed landslide costs far more than a false alarm, so the operating point targets at least 90% detection, with a margin: the Watch threshold is the highest one whose bootstrap 5th-percentile out-of-fold recall is still at least 90%.
- **Frozen test set, single look.** The 2018 test year was frozen with checksums before modelling and evaluated once with the model locked (SHA-256 freeze record); a second run needs an explicit, logged flag.
- **Never silent when blind.** If sensors are untrusted or sources disagree and any independent source shows heavy rain, the level is raised, using the *maximum* of the sources and never the minimum.
- **Missing data is explicit.** Missing values are NaN (never zero) and counted; LightGBM handles NaN natively.
- **Causal features.** Every feature at day D uses information up to and including day D only; a test perturbs the future and checks nothing changes.
- **No label leakage.** Latitude, longitude, cell id, region, month, rain-source and terrain-source columns are metadata and can never be model inputs; station presence is excluded because inventories cluster where gauges and people are.
- **Repaired values never feed the hazard model's rain features.** Repair under-estimates heavy rain (only 42.5% of repaired values on 50+ mm days reach half the truth), so it is used only for the integrity summary.

3.5 Software structure and compute

Python 3.13 with a project-local virtual environment; pandas 3.0, NumPy 2.5, LightGBM 4.7, scikit-learn 1.9, FastAPI, Streamlit. The code is a package `aetherguard` with sub-packages `audit`, `events`, `splits`, `ingest`, `features`, `integrity`, `corrector`, `hazard`, `cases`, `alerts`, `serve`, `pipeline`, `finalreport`, YAML configuration in `configs/`, 1,323 tests (offline and hermetic; real-data tests are opt-in through markers) and a documented design file per phase in `docs/`. The target machine has Windows 11, 16 GB RAM and no GPU, so only CPU-friendly models (gradient-boosted trees, linear models) are used; the largest download is CHIRPS at 14.97 GB.

4 Datasets

4.1 What was used

Dataset	Provider	Contents	Coverage used	Volume	Role
GHCN-Daily PRCP	NOAA NCEI	Daily rain-gauge series with quality flags	140 stations (138 India, 2 Nepal), 1990-2024; 2007-2024 for the engine	3,099,630 valid values in the audit; 26.5% of station-days populated	Stage-1 training and test; corrector target; station trust
IMD gridded rainfall	India Meteorological Dept. (imdlib)	Gauge-based daily rain grid, India only	0.25°, 129 × 135 grid, 2007-2024; 4,964 well-covered points	18 yearly files, ≈ 25 MB each	Primary rain source; day basis reference
CHIRPS v2.0 daily	UCSB Climate Hazards Center	Satellite + gauge rain estimate	0.05°; 4,566 days (2007-2018, 2023, 1 Jun-15 Aug 2024); UTC days	14.97 GB raw; cropped store 560 × 520 px, float16	Independent second source; only source for Nepal/Bhutan cells
NASA POWER / MERRA-2	NASA	Soil wetness (surface GWETTOP, root zone GWETROOT)	≈ 0.5° × 0.625°, 2007-2024 (6,575 days), 59 × 43 points	Small	Antecedent wetness features
Copernicus GLO-90 DEM	ESA / Copernicus	Elevation model	90 m; 495 tiles (23 ocean tiles absent)	2.32 GB	Slope, relief, curvature, aspect, elevation
NASA Global Landslide Catalog	NASA	Legacy landslide records	11,033 rows, 1988-2017	—	Labels
COOLR reports	NASA	Report-based inventory with location accuracy	14,750 rows, 1974-2025	—	Labels
COOLR events	NASA	Mapped landslide inventories	40,154 points, 2011-2024	—	Labels (inventory-window records)
HMA landslide catalog v2	NSIDC	High Mountain Asia point catalog	2,801 points, 1990-2018	—	Labels
Natural Earth admin-0	Natural Earth	Country polygons (50 m)	Global	—	Country / region assignment
ERA5 hourly (Open-Meteo)	ECMWF / Copernicus	Reanalysis hourly rain	4 points (Wayanad + 3 Kerala), 8 API calls	—	Case-study day-basis checks

All sources are open and need no login except the NASA/NSIDC label catalogs (Earthdata). Each downloaded file is recorded in a checksummed manifest with its URL and licence.

4.2 Landslide labels: how the positives were built

Four catalogs were merged into one label store of **68,738 records** with a uniform schema. Rules, all fixed before modelling:

- **Region by country, not by box.** Himalaya = India, Nepal or Bhutan inside lat 22-36°N, lon 72-98°E; Western Ghats = India inside lat 8-21.5°N, lon 72.5-78.5°E. The coarse box also contains Myanmar (6,823 automated events dated 2015), Bangladesh (715 events in 2023), Pakistan and China, which must not be counted.
- **Positives need a rain-class trigger** (text containing rain, downpour, monsoon or cyclone). Flooding is deliberately not rain-class; earthquake, construction and unknown triggers are excluded from positives but kept as buffer records.
- **Location accuracy ≤ 5 km** and a date within 2007-01-01 to 2024-12-31. Duplicates across catalogs are removed by priority (HMA > COOLR reports > GLC > COOLR events).
- **Day-exact versus inventory-window.** COOLR event polygons carry only an inventory headline date. They are never day-exact positives; the 2023 India inventory (176 events in 24 cells, window 16 Jun-31 Jul 2023) is used only as a window-level test.
- The result: **919 day-exact positive cell-days** (from 956 events; 780 Himalaya and 176 Ghats events) plus the 176 window events.

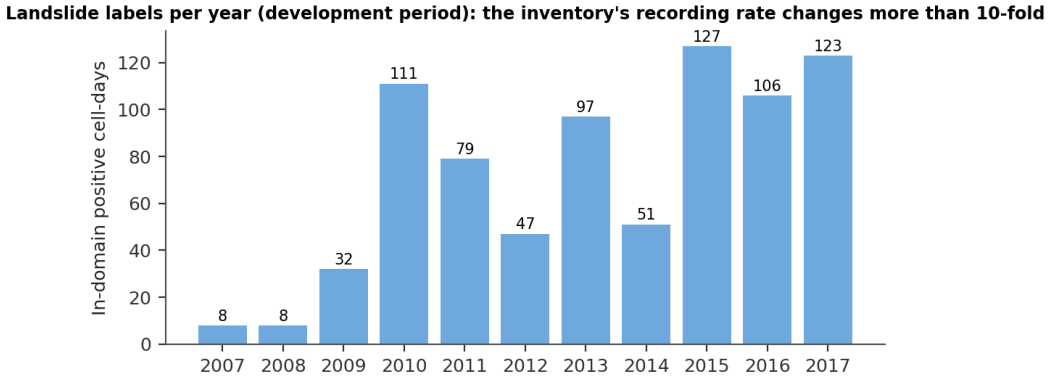


Figure 3. In-domain positive cell-days per year in the development period (2007-2017). Source: model card, base_rates.

4.3 Frozen splits

Split	Dates	Cell-days	Events	Use
train	2007-2015	591	616	Development (training and cross-validation)
validation	2016-2017	233	242	Development (pooled with train for year-blocked cross-validation)
test_2018	2018	85 (80 in the model's domain)	87	Single frozen evaluation; contains the Kerala floods
test_sparse	2019-2024	10	11	Reporting only (too few; no rain look-back)
window_2023	16 Jun-31 Jul 2023	24 cells	176	Window-level detection test (inventory is partial)

Splits were written to read-only Parquet files with SHA-256 checksums; integrity tests assert that no (cell, date) appears in two splits and that checksums still match. Kerala 2018 (25 cell-days; 23 inside the case-study grid) is a case study inside test_2018, and Wayanad 2024 (0 labelled positives) is a never-tuned qualitative case study.

4.4 Grid, domain and the day basis

The model unit is a **0.05° cell and a calendar day** (about 5 km), anchored on the CHIRPS lattice. The domain (lat 8-36°N, lon 72-98°E) holds 166,728 terrain cells, of which 69,464 are in the Himalaya region, 21,826 in the Western Ghats and 75,438 out of scope (other countries, sea).

Which day does a rain value belong to? IMD's day runs 08:30 IST to 08:30 IST; CHIRPS, MERRA-2 and ERA5 use UTC days. A landslide at 02:00 IST on date D is captured by IMD day D but by UTC day D-1. To make products comparable, UTC products are converted to the IMD day: $P_{IMD}(D) = 0.875 \times P_{UTC}(D-1) + 0.125 \times P_{UTC}(D)$. Against exact hourly ERA5 at Wayanad this approximation has a mean absolute error of 1.85 mm, no bias, and 7.7% error on heavy days (a single day can be off by about 10 mm). The assumption was tested rather than taken on faith (Figure 4): gauge-vs-IMD correlation peaks at lag 0 (median 0.69, all 11 stations), gauge-vs-CHIRPS(UTC) peaks at lag -1, and the realigned CHIRPS peaks at lag 0.

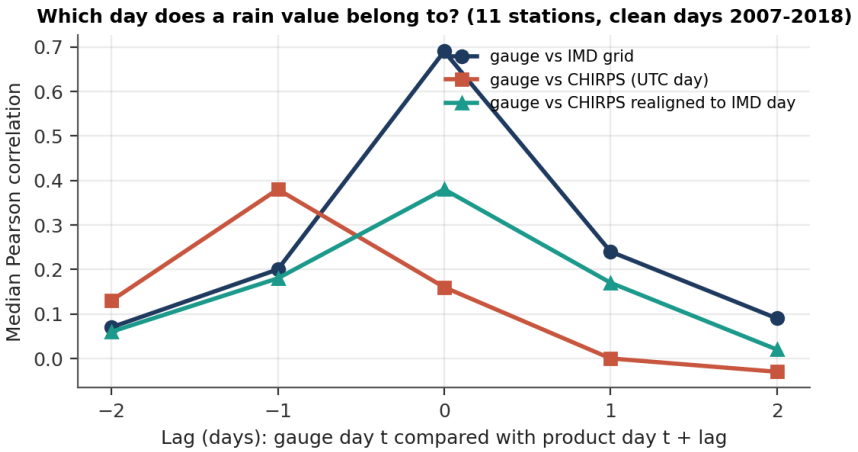


Figure 4. Lag correlation of clean gauge days with IMD and CHIRPS (medians over 11 stations; source: reports/phase2b_day_basis.json). IMD agreement shows label consistency, not independence, because the grid probably ingests some of the same gauges.

4.5 Negatives and the 'universe' sample

A model that only sees landslide days cannot learn when landslides do not happen, and a sample that is half landslides cannot report an honest false-alarm rate. Two kinds of non-event rows were therefore built, seeded and frozen:

- **Matched negatives** (10 per positive, per split and region): same region and same calendar months as the positives, at least 3 cells (0.15°) and ±3 days from *any* event record in the store (positive or not), drawn among cells whose mean slope is at least the 5th percentile of training-positive cells (2.544°). These train the model.
- **Universe sample** (200,000 uniform draws per split, 100,000 per region): random cell-days from all candidate cells and all days, each carrying a weight (universe size ÷ sample size) so that weighted counts represent **all** cell-days. These measure false-alarm shares and set thresholds. Rows near an event record are flagged ambiguous and excluded from false-alarm counting (sensitivity with them included is also reported).
- Development numbers: 789 in-domain positives (35 more fall below the slope floor and are counted as misses), 8,240 matched negatives, 399,272 universe rows representing 146.7 million cell-days; the base rate is 0.54 landslide cell-days per 100,000.

4.6 The feature store

Group	Count	Contents	In the frozen hazard model?
rain_	14	Accumulations over 1, 3, 5, 7, 14, 30 days (IMD day basis, IMD primary else CHIRPS aligned); max 1-day rain in 7/30 days; wet days in 30; days since 10 mm; 1-day share of 3-day total; 3/7/30-day totals relative to the cell's own 2007-2015 climatology	Yes (14)
src_	13	IMD and CHIRPS accumulations separately and their disagreement (difference, ratio, both-valid flag)	No: not selected by cross-validation (within noise); disagreement is handled by the alert engine
wet_	6	MERRA-2 surface and root-zone wetness from UTC day D-1 (so nothing after the rain window is used) and their 3- and 7-day changes; ocean points masked	Yes (6)
stat_	10	Mean/std elevation, relief, mean/90th-percentile/max slope, curvature, aspect (sin, cos), wetness-index proxy from the 90 m DEM	Yes (10)
integ_ / corr_	5 / 4	Station-trust summaries near the cell; corrector quantiles for the cell	Tested, not selected (no gain in cross-validation or on test)

The frozen model therefore uses 30 inputs: 14 rain + 6 wetness + 10 terrain. 17 of them carry a monotone constraint.

4.7 Data-quality findings that shaped the design

- Post-2018 labels are almost absent (the Western Ghats have none in the mapped inventories; Wayanad 2024 is in no NASA catalog), so the test set is 2018 plus a window-level 2023 test.
- The inventory's recording rate changed more than 10-fold across 2007-2017, making absolute probabilities unreliable but ranking usable.
- IMD's grid is India-only: Kathmandu and Pokhara are missing on every day, so Nepal/Bhutan cells depend on CHIRPS.
- MERRA-2 surface wetness reads exactly 1.0 over water (25.4% of values); it is masked wherever root-zone wetness is missing.
- Gauge records are sparse: only 26.5% of station-days have a value, clean stretches are short (median 6 days), and just 10 real quality flags exist in 2007-2024, so faults had to be injected into clean real series to train and score Stage 1.
- The neighbour-gauge design (5 neighbours within 150 km) is almost unusable: only 5.1% of clean days have two valid neighbours; a wider setting (10 neighbours, 300 km) reaches 78.4% and is used.

5 The models

5.0 Which model was built on which data

Model	Algorithm	Trained on	Chosen / calibrated on	Evaluated on	Output
L1 rules	Deterministic checks (bounds, stuck, duplicate block, gaps)	No training; thresholds measured on 231,539 real clean station-days	same measurement	TEST stations	Flag + reason
Stage-1 engine	LightGBM, 7 classes (ok + 6 fault types); isotonic calibration; arbiter; repair	Faults injected into real clean gauge series: 84 TRAIN stations (5 injection seeds)	28 VAL stations (hyper-parameters, calibration, threshold)	28 TEST stations (11 with episodes), unseen	Trust, fault type, genuine extreme, repaired value
Extreme-rain corrector	LightGBM quantile regression (0.5, 0.9, 0.99) + exceedance classifier	Gauge rain vs CHIRPS-side predictors at 22 TRAIN stations (20,980 days)	8 VAL stations (7,808 days)	6 TEST stations (6,143 days)	Gauge-like rain quantiles
Hazard model (Stage 2)	LightGBM binary, 30 inputs, monotone constraints, 20-member ensemble, isotonic calibration	2007-2017: 789 positives, 8,240 matched + 20,000 universe negatives	7-fold year-blocked cross-validation (all choices out-of-fold)	Unseen 2018: 80 positives + 200,000-row universe; window 2023; Kerala 2018; Wayanad 2024	Calibrated risk, interval, top-5 reasons
Baselines	Intensity-duration rule; LHASA-style rule; logistic regression	Same development rows (fitted inside each fold)	Their own out-of-fold thresholds	Same sets as the model	Scores
Alert engine	Rules on top of the model	Percentiles from the 2007-2017 development universe only	Thresholds from the model card	2018 universe, window, cases	Level, confidence, reasons

5.1 Stage 1: the sensor-integrity engine

Problem. Real gauge faults are too rare to learn from (391 NOAA-flagged values in the whole history of the 140 stations, only 10 in 2007-2024, mostly whole duplicated months in 1930-66). So faults were **injected into real clean series**, with magnitudes calibrated against the real flagged faults, and results on injected faults are reported separately from the 10 real flags.

Fault type	What it simulates	Injection (documented ranges)	Episodes TRAIN / VAL / TEST
stuck_value	A constant positive value repeats	6-30 days at a positive value from the station's 50th-90th wet-day percentile (floor above the longest natural streak, 5 days)	71 / 27 / 22
zero_fault	Clogged or dead gauge reading zero while it rains	3-30 days of zeros	76 / 29 / 21
spike	Implausibly large single-day value	×10 to ×50 of a local wet value, or 500-1500 mm	67 / 29 / 23
scale_drift	Multiplicative bias drifting over weeks	30-90 days, factor moving linearly from 1 to 0.3 or 2.5	31 / 12 / 14 (short of the planned minimum)
offset_noise	Small positive readings on dry days	10-60 days adding U(0.2, 1.0) mm on dry days	58 / 22 / 19
duplicate_block	A block copied from earlier days (NOAA flag D)	7-31 days copied from an earlier fully observed block	67 / 28 / 23

Episodes are placed only on clean stretches (valid values, no NOAA flag, station-year with at least 150 valid days in the engine dataset), never overlap, and keep at least 5 days apart. Stations are split by station id (84 train / 28 validation / 28 test, fixed seed 20260930), so evaluation is on gauges the model has never seen.

The four layers

- **L1 deterministic rules:** value below 0 or above 1,500 mm/day; at least 3 identical positive consecutive values (false-flag rate on 231,539 real clean days: 0.075%); a block of at least 7 days equal to an earlier block with at least 3 positive days; gaps reported with run length. Each threshold's false-flag rate was measured on real clean days.
- **Learned model:** LightGBM with 7 classes on 63 causal features in four groups: *own* (26: trailing 7/14/30-day mean, std, zero fraction, max, distinct values, run lengths, wet-day fraction, ratios to the station's own 2007-2015 percentiles), *nbr* (9: neighbour-gauge reference, residuals, correlation), *grid*

(24: differences, ratios, correlations and sums versus IMD and CHIRPS-aligned; agreement count of independent references) and *wet* (4: soil wetness). Training uses five injection seeds, and a 'reference dropout' augmentation that blanks each reference group on 30% of rows so the model tolerates missing references. Hyper-parameters (4-candidate grid) and the calibrator are chosen on validation stations.

- **Calibration and threshold:** isotonic regression gives a probability of fault; the threshold is chosen on the *combined* decision (L1 or model, after arbiter) on validation clean days so that false flags are at most 1% (also 0.5% and 2% reported).
- **Arbiter (genuine-extreme protection):** a flagged reading of at least 20 mm is retained as a real extreme when at least two independent references among IMD, CHIRPS-aligned and the wide-neighbour mean are at least 30% of the reading (with one reference: at least 60% and fault probability below 0.9). Suspicious readings that cannot be confirmed stay in the data marked low-trust, never deleted.
- **Repair:** a weighted median of the available references (weights from inverse validation error) with a 10-90% band from empirical validation residuals; the raw reading is always preserved.

Out-of-fold outputs (5-fold, station-grouped) were produced for every station-day 2007-2024 (920,500 rows; trust is NaN on 76.9% where no value was reported); the hazard side only ever sees these out-of-fold values.

5.2 The extreme-rain corrector

Target: gauge daily rain on clean, unflagged days. Predictors: CHIRPS realigned to the IMD day (day and trailing 3-day), soil wetness (D-1), terrain of the station's cell, station elevation, season. **IMD is deliberately excluded** from the headline corrector because the IMD grid probably ingests the same gauges (a leakage-risk variant with IMD is reported only for comparison). The models are LightGBM quantile regressions at 0.5, 0.9 and 0.99 with an upper-quantile loss so extremes are not pulled toward the mean, plus a classifier for 'exceeds the station's own 99th wet-day percentile'.

5.3 Stage 2: the landslide hazard model

Model. LightGBM binary classifier on 30 inputs with **monotone constraints** (more rain, steeper slope, greater relief, or more wetting never lowers risk; more days since a 10 mm day never raises it). Frozen configuration: 15 leaves, minimum 40 rows per leaf, feature fraction 0.5, no L2, learning rate 0.05, 118 trees, matched negatives plus 20,000 universe negatives (weighted to contribute equally). A 20-member bootstrap ensemble gives a 10-90% interval; SHAP values from LightGBM give the top-5 reasons per alert.

Protocol, in two versions (disclosed). The first version (v1) used the 2016-2017 validation split for everything: choosing among 48 hyper-parameter candidates, early stopping, calibration and the 90%-recall threshold. That made the threshold optimistic: on the unseen year 2018 it reached only 78.7% recall (target 90%), calibration over-predicted by 36%, and the model only tied the baselines. Because of that, protocol **v2** was adopted (and the dependence on the v1 test result is stated plainly): all choices are made by **7-fold year-blocked cross-validation over 2007-2017** (folds 2007-08, 2009-10, 2011-12, 2013-14, 2015, 2016, 2017); calibration and thresholds come from pooled out-of-fold scores; the final model is refit on all development rows; baselines get the same treatment; and 2018 is evaluated exactly once after everything is frozen.

Feature-group candidate (selected hyper-parameters)	Inputs	Cross-validated false-alarm share at 90% recall [± SE]	Decision
full	43	19.8% ± 1.4	
stat without elevation columns	41	24.5% ± 1.7	Worse by 4.7 points (> SE): elevation kept
drop src (frozen model)	**30**	**18.8% ± 1.1**	**Selected (simplest within noise)**
drop wet	37	20.2% ± 1.4	
drop src + integ	35	20.9% ± 1.7	No gain
drop src + corr	34	19.7% ± 1.7	No gain
drop src + integ + corr	39	19.3% ± 1.4	No gain

The 17-candidate hyper-parameter grid spans 21.8-24.4% before group selection, i.e. within about one standard error: the exact settings matter little.

Thresholds and alert levels (from pooled out-of-fold scores)

Level	Rule	Calibrated risk (per cell-day)	Out-of-fold recall	Out-of-fold false-alarm share
Watch (recall-first)	Highest threshold whose bootstrap 5th-percentile recall ≥ 90%	4.13e-06	91.8%	20.95%
Warning	Out-of-fold recall 75%	1.58e-05	75.0%	9.05%
Emergency	Out-of-fold recall 50%	3.70e-05	50.1%	3.44%

Baselines (identical data, identical protocol)

- **Intensity-duration threshold:** alert when rain over N days (1, 3, 5, 7, 14, 30) exceeds the 10th percentile of training positives' N-day rain; score = the largest ratio to those thresholds.
- **LHASA-style:** five slope-quantile susceptibility classes × rain trigger (3-day rain relative to the cell-month mean); score = class × ratio.
- **Logistic regression:** standardised 1/3/7/30-day rain, 3-day anomaly, soil wetness and mean slope, class-weighted.

5.4 The alert engine

- **Levels** Normal / Watch / Warning / Emergency from the out-of-fold thresholds. An **uncertain_high** rule raises Watch when the upper bound of the 10-90% interval crosses the Watch threshold while the point estimate does not (recall-first).
- **Hysteresis:** a cell steps up immediately but steps down one level only after 2 consecutive days below 0.8 × the current level's threshold; this halves flapping (23.2 → 12.7 flaps per 1,000 cell-days in the 2023 window).
- **Data confidence** (High / Degraded / Insufficient) from rain-source availability, IMD-vs-CHIRPS disagreement outside the development 5-95% range, flagged or untrusted nearby stations, and the station out-of-distribution guard.
- **Never silent when blind:** if confidence is Degraded or Insufficient and any independent source shows heavy rain (derived at the 95th percentile of the development universe: 32.5 mm in one day or 79 mm in three), the level is raised to at least Watch, using the maximum of the sources.
- **Out-of-distribution station guard** (the remedy for the Nepal shortfall): a station whose trailing 90-day flag rate exceeds 3 × the engine target is marked 'uncertain' (data confidence Degraded), never silently 'faulty'.
- **Reasons** with every alert: top-5 SHAP factors, rain-source spread (IMD, CHIRPS, station), sensor-trust summary, confidence, interval; plus a cost-loss report.

5.5 Serving and interfaces

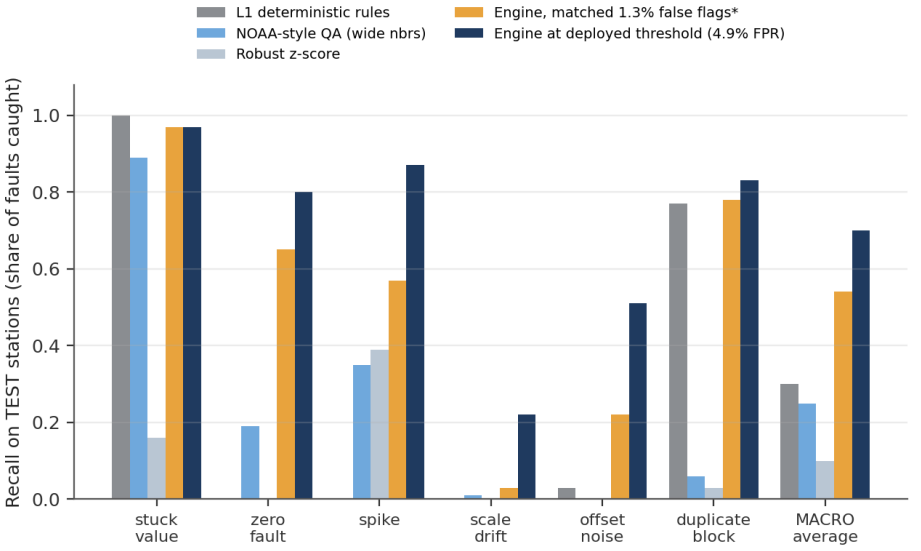
FastAPI (binds 127.0.0.1): `/health` (model-freeze status, card hash, store status), `/risk` (cells for a date, region or bounding box), `/risk/cell/{cell_id}` (full explanation), `/sensors` (station trust and fault type), `/alerts` (alert list). **Streamlit dashboard** with six pages: risk map, why-this-alert, sensor trust, Wayanad 2024 replay, Kerala 2018 replay, evaluation. A replay store holds precomputed per-day alert tables; live mode is out of the validated pipeline and clearly labelled.

6 Results and accuracy

6.1 How to read the numbers

Term	Meaning
Recall / detection rate	Share of real events (landslide cell-days, or injected faults) that were flagged. 85% means 68 of 80.
False-alarm share	Share of all cell-days that were put on alert, measured on the weighted random universe. Lower is better; it is an upper bound because the inventory is incomplete.
95% interval [lo, hi]	Bootstrap interval: events are resampled as events, random cell-days iid, gauges as stations. A wide interval means the data cannot separate two numbers.
Out-of-fold (OOF)	Each row was scored by a model that never saw its year (or station): an honest estimate inside the development period.
Frozen / single look	Model and thresholds fixed and checksummed before the test year was scored once.

6.2 Stage 1: how well sensor faults are caught



*oracle-matched: threshold picked on TEST labels so the false-flag rate equals NOAA-wide's 1.3%; not deployable.

Figure 5. Recall per injected fault type on the 28 unseen TEST gauges (11 carry episodes). Source: reports/phase2_integrity_report.json.

Method (TEST gauges)	stuck	zero	spike	drift	offset	dup.	macro	false-flag rate
Range check	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.000
L1 deterministic rules	1.00	0.00	0.00	0.00	0.03	0.77	0.30	0.002
NOAA-style QA, design neighbours	0.88	0.03	0.26	0.00	0.00	0.00	0.20	0.001
NOAA-style QA, wide neighbours	0.89	0.19	0.35	0.01	0.00	0.06	0.25	0.013
Robust z-score (1%)	0.16	0.00	0.39	0.00	0.00	0.03	0.10	0.01
Isolation forest (1%)	0.00	0.00	0.22	0.00	0.00	0.00	0.04	0.01
Engine, model score only (1%)	0.84	0.63	0.57	0.02	0.19	0.02	0.38	0.01
Engine, combined (VAL-chosen 1% threshold)	**0.97**	**0.80**	**0.87**	**0.22**	**0.51**	**0.83**	**0.70**	**0.049**
Engine, matched to NOAA-wide rate (oracle, not deployable)	0.97	0.65	0.57	0.03	0.22	0.78	0.54	0.013

Reading it. The engine is clearly better on zero-reading, offset and spike faults and ties or loses to simple rules on stuck-value and duplicate-block faults (L1 already catches those almost perfectly; the engine's gain there comes from L1 inside it). **Slow drift is essentially undetected by every method** (0.22 for the engine at 4.9% false flags, 0.03 at a matched rate); only 14 test episodes exist. **Shortfall:** the validation-chosen threshold gave 1.0% false flags on validation but **4.9% on test**. 91% of the test false

flags (496 of 545) come from the two Nepal gauges, which hold 51% of test clean days and look unlike the Indian training stations; without them the test rate is 0.9% (macro recall 0.72, post hoc). It was not retrained after seeing test; the remedy is the out-of-distribution guard (§5.4).

Other Stage-1 results	Value
Genuine-extreme retention, TEST (retained / extremes): strict / wide / one-neighbour / uncorroborated proxy	3/3 · 9/9 · 17/18 · 24/26 (the robust z-score retains only 21/26 on the proxy). No single number is the safety result.
Cost of the arbiter in injected-spike recall	None: 20 of 23 spikes flagged with and without it; it retained 0 injected spikes.
Repair accuracy on flagged faulty days (TEST, 688-day common set)	MAE 11.24 mm vs previous-day 19.44, neighbour mean 15.34, zero-fill 13.17, unrepaired 27.71. 10-90% band covers 75.5%.
Repair on heavy rain (≥ 50 mm days)	Only 42.5% of repaired values reach half the truth (40 days): repair under-estimates, so repaired values never feed the hazard model.
Real NOAA flags in 2007-2024 (n = 10, too few to rank)	Engine flags 9 of 10 at 1.0% false flags on unflagged days; NOAA-style QA 6; L1 0; robust z-score 1.
Ablations (model-score TEST macro recall, full = 0.377)	own features only 0.255; without wetness 0.333; without extra injection seeds 0.277; without gridded or neighbour features 0.404 / 0.400 (no help on TEST).
Stability over three other injection seeds (combined engine)	Macro recall mean 0.679 (0.649-0.702).

6.3 The extreme-rain corrector

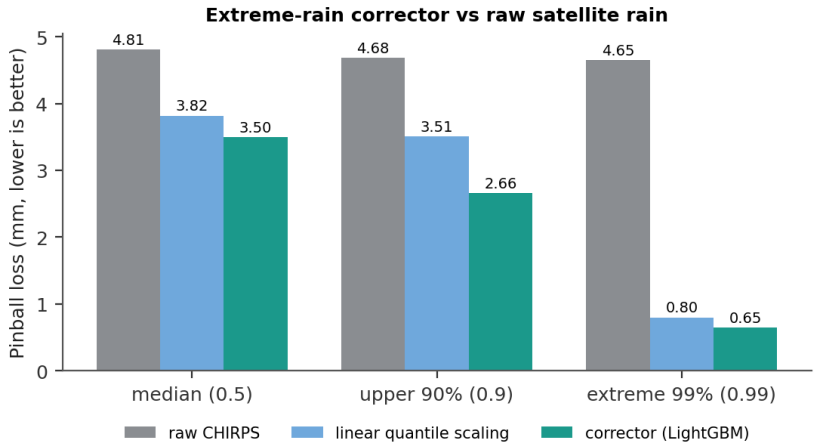


Figure 6. Pinball loss (lower is better) on TEST gauges, CHIRPS-only predictors. Source: reports/phase3_corrector_report.json.

- The corrector beats raw CHIRPS at all three quantiles with paired intervals excluding zero (e.g. -2.02 mm [-3.41, -0.52] at the 90% quantile) and beats linear quantile scaling too.
- Quantile coverage on TEST: the 0.9 quantile covers 90.2% and the 0.99 quantile 98.7% of gauge days, versus 64.1% for raw CHIRPS.
- Bias on the top 1% of gauge days (≥ 72.9 mm): raw CHIRPS -66.6 mm [-84, -59]; the 99% quantile +4.8 mm [-14, +12] (unbiased); the median and 90% quantile still under-predict strongly.
- Exceedance of the station 99th percentile (33 test days): PR-AUC 0.071 [0.045, 0.253] vs 0.047 [0.031, 0.154] for raw CHIRPS: not separable.
- **Wayanad 30 Jul 2024:** predicted q50 31 mm, q90 71 mm, q99 127 mm versus IMD 179 mm: it raises the estimate over CHIRPS but **does not close the gap** (target: q90 ≥ 50% of IMD, achieved 40%). Training gauges are mostly lowland, so transfer to windward slopes is not guaranteed.
- Only 6 test stations and 63 top-1% rows exist, so every interval is wide.

6.4 The landslide hazard model

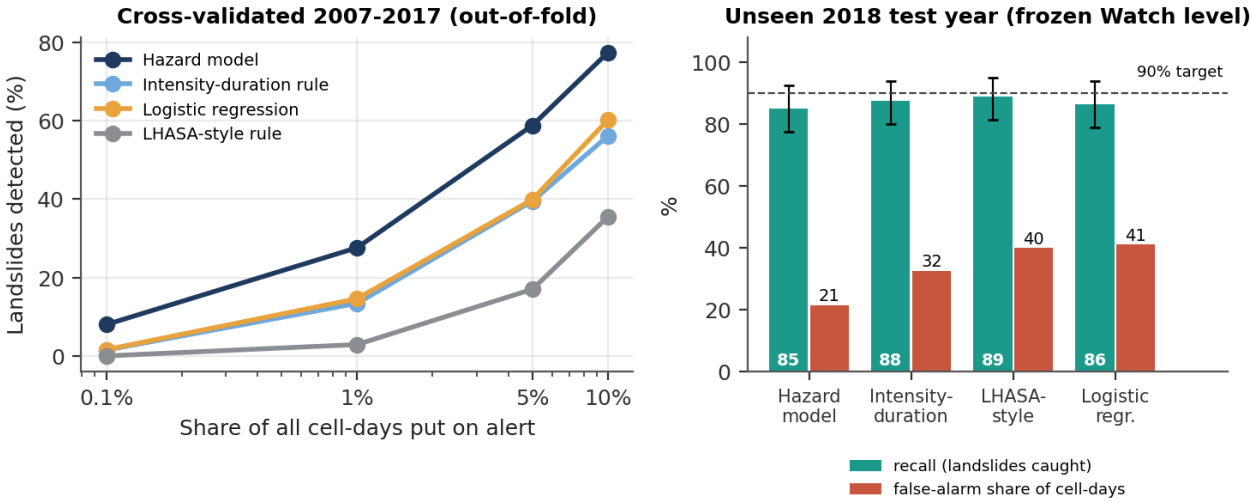


Figure 7. Left: detection rate at fixed false-alarm shares, out-of-fold 2007-2017. Right: unseen 2018 at each scorer's frozen Watch threshold (bars: recall with 95% interval; red: false-alarm share). Sources: reports/phase3_hazard_cv_report.json and phase3_hazard_report.json.

Cross-validated, 2007-2017 (out-of-fold, same protocol for every scorer)

Scorer	False-alarm share at 90% recall [95% CI]	Recall at 1% FA	Recall at 5% FA	Paired gap to model [95% CI]
Hazard model	**18.8%** [17.3, 21.3]	27.6%	58.8%	—
Intensity-duration	32.1% [30.8, 34.8]	13.4%	39.5%	+13.3 points [11.2, 15.3]
Logistic regression	36.1% [31.5, 41.2]	14.6%	39.9%	+17.3 points [12.7, 21.6]
LHASA-style	37.7% [35.1, 41.4]	2.9%	17.0%	+18.8 points [15.8, 21.9]

Per-fold false-alarm share ranged 15.0-22.3%, so the result is stable across years, though the 2007-08 fold has only 16 positives.

Unseen 2018 (one frozen evaluation)

Scorer at its frozen Watch threshold	Recall [95% CI]	False-alarm share [95% CI]	FA gap vs model (baseline minus model) [95% CI]	Recall gap (model minus baseline) [95% CI]
Hazard model	**85.0%** [77.5, 92.5]	**21.4%** [21.2, 21.6]	—	—
Intensity-duration	87.5% [80.0, 93.8]	32.4% [32.2, 32.7]	+11.0 [10.8, 11.2]	-2.5 [-8.8, 3.8] (tie)
LHASA-style	88.8% [81.2, 95.0]	39.8% [39.6, 40.1]	+18.4 [18.2, 18.7]	-3.8 [-12.5, 3.8] (tie)
Logistic regression	86.2% [78.8, 93.8]	40.9% [40.6, 41.1]	+19.5 [19.2, 19.7]	-1.3 [-6.3, 3.8] (tie)

Recall-first gate: not reached. At the frozen Watch threshold 85.0% [77.5, 92.5] of the 80 in-domain 2018 events were caught (target 90%, shortfall 5.0 points; the interval still contains 90%). At the plain-90% threshold: 82.5% recall at 19.2% false alarms. Counting the 5 events below the slope floor as misses, overall recall is 80.0% [71.8, 88.2]. Warning and Emergency overshoot their out-of-fold targets (realised recall 65.0% and 38.8% vs 75% and 50%).

Verdict versus baselines. At equal recall (thresholds re-picked on test labels, descriptive only) the model ties the intensity-duration and LHASA-style rules and beats logistic regression. At the frozen operating points the baselines catch about as many events (differences within noise) but alert on 11-19 points more cell-days. With only 80 positives, test_2018 cannot separate the scorers on recall.

Per region, calibration, window test, leave-region-out, ablations

Check	Result
Per region (2018, Watch threshold)	Himalaya (48 events): recall 87.5% [77.1, 95.8], false alarms 23.9%. Western Ghats (32 events): recall 81.2% [65.6, 93.8], false alarms 12.6%.
Calibration on test (per 100,000 cell-days)	All: predicted 0.522 vs observed 0.596 (calibration error 12.5% of the base rate). Himalaya over-predicted (0.573 vs 0.458); Ghats under-predicted about 3× (0.341 vs 1.089). Cross-validated per-year observed rates varied more than 10×, so probabilities are uncertain by about a factor of two even where the ranking is good.
2023 India window (24 cells, Watch threshold)	All 24 cells detected (cell or 8 neighbours alerting on any day of the window), by every scorer. Control cells alerting on at least one day: model 81.8%, logistic 99.1%, intensity-duration and LHASA-style 100%. Share of control cell-days alerted: model 46.7% vs 68-72%. False alarms are an upper bound (inventory partial).
Leave-region-out, Himalaya → Ghats (32 events)	False-alarm share at 90% recall: full model 29.5% [8.6, 38.5]; Himalaya-only 23.1%; without elevation 20.9% — ties.
Leave-region-out, Ghats → Himalaya (48 events)	Full model 28.8% [16.7, 43.0] vs Ghats-only model 53.7% [32.3, 70.9] — the full model is better (paired +24.9 points [11.9, 32.0]); transfer from the Ghats to the Himalaya fails regardless of elevation.
Elevation as a location proxy?	Dropping elevation does not change transfer in either direction, so this check does not show elevation acting as a proxy; but dropping it costs 5.9 points of false alarms on test (27.3% vs 21.4%) at about equal recall. Both checks have fewer than 100 positives.
Ablations on test (own recall-first thresholds)	frozen model: recall 85.0%, false alarms 21.4%; full 83.8% / 21.6%; drop wetness 83.8% / 22.9%; +integ 85.0% / 23.4%; +corr 83.8% / 23.0%; +integ+corr 82.5% / 22.4%. None is better than the frozen model.

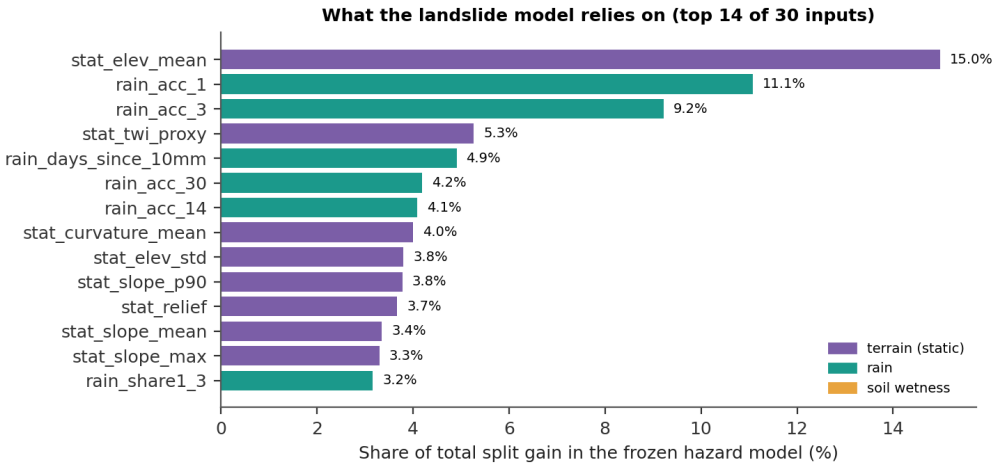


Figure 8. Share of total split gain per input in the frozen hazard model (source: model card). Elevation is the single largest input; rain over 1 and 3 days follows.

Monotonicity: all 17 constrained inputs of the final model held on a partial-dependence grid with zero violations. Model history: the first protocol (v1) is kept in `reports/phase3_hazard_report_v1.md`; its cross-validation-free validation numbers favoured the model by construction, and its test numbers (recall 78.7% [70.0, 87.5] at the validation threshold, ties with all baselines) triggered protocol v2.

6.5 The alert system and the case studies

The complete system adds the uncertainty rule, blind-sensor rule and hysteresis on top of the model. Results on 2018, using the frozen model and thresholds:

Level	Alerts per 1,000 cell-days	Recall in domain (of 80)	Recall incl. out-of-domain misses (of 85)	Detected same day
Watch or higher	271.6	91.2%	85.9%	73
Warning or higher	91.7	70.0%	65.9%	56
Emergency	35.4	38.8%	36.5%	31

What each layer adds at Watch	Alerts per 1,000 cell-days	Recall in domain
Point risk (model alone)	218.0	86.3%
+ uncertain_high rule	266.5	91.2%
+ blind-sensor rule	271.1	91.2%
+ hysteresis	271.6	91.2%

So the step from 86% to 91% recall comes entirely from the uncertainty rule, at a price of about 48 extra alerts per 1,000 cell-days. The blind-sensor rule raised 1,300 cell-days in the 2018 universe, none near an event: it has not yet rescued a real event in these data (every labelled event was already caught by the model).

Case studies (never trained or tuned on)

Case	Result
Kerala 2018 (1-20 Aug; 8,048 cells; 23 labelled cell-days)	**23 of 23 detected** at Watch or higher (20 at Warning+, 14 at Emergency); median first alert 7 days ahead, limited by the 8 days of history in the case table. Watch+ covered on average 45% of cells (peak 67% on 17 Aug); flaps 10.3 vs 22.2 per 1,000 cell-days without hysteresis.
2023 India window	24 of 24 cells detected; however 60% of other cell-days reached Watch+ and 91% of other cells alerted on some day (an upper bound because the inventory is partial).
Wayanad 2024 (15 Jul-5 Aug; 140 cells; no label)	**Limitation: the levels saturate.** The focus cell 0.8 km from Mundakkai is at Emergency on all 22 days; on average 35% of the 140 cells are at Emergency. The risk takes only 10 distinct values; its maximum (4.5e-04) is shared by 17 July and 30 July, so the disaster day is joint-first but not singled out. Not tuned away.

Cost-loss. The thresholds are optimal only if acting costs at most 4.1e-06, 1.6e-05 and 3.7e-05 of the loss avoided (relative expense 0.43, 0.54, 0.82 — below 1 beats both 'never alert' and 'always alert'). At cost-loss ratios of 1/10, 1/50 or 1/200 the cost-optimal rule is effectively never to alert: almost no cell-day reaches a risk that high, so the levels are defensible only where acting is very cheap relative to the loss avoided.

6.6 Scorecard against the original goals

Goal (design spec)	Target	Achieved	Status
Detect rain-triggered landslide days (hazard model alone, 2018)	≥ 90%	85.0% [77.5, 92.5]	Not reached
Same for the complete alert system (2018)	≥ 90%	91.2% (73/80)	Reached, at 27% of cell-days on Watch+
Fewer false alarms than standard methods	Beat the 3 baselines	21.4% vs 32.4-40.9% at similar recall (paired, clear)	Reached
Detect sensor faults better than NOAA-style QA	Beat baselines	Macro 0.70 vs 0.25-0.30 (4.9% false flags); 0.54 at matched rate	Reached on zero/offset/spike; drift unsolved
Retain genuine extremes	≈ 100% of corroborated extremes	3/3, 9/9, 17/18, 24/26 (small denominators)	Reached, weakly measurable
Extreme-rain corrector closes the Wayanad gap	q90 ≥ 50% of IMD	40% (71 mm vs 179 mm)	Not reached
Calibrated probabilities	Calibrated	Test error 12.5% of base rate; ~2× year-to-year uncertainty	Partly
Never silent when blind	Rule exists and is tested	Implemented; 0 rescued events so far	Reached (not yet proven useful)
Frozen, single-look evaluation with intervals	Required	Done; one disclosed bug-fix rerun	Reached

7 How AetherGuard addresses the failures

Failure of traditional systems	What AetherGuard does	Evidence it works	What remains
F1 Rain products under-measure extremes	Two independent rain sources aligned to the same day (IMD primary, CHIRPS second); a quantile corrector; the alert engine takes the *maximum* of sources and escalates when sources disagree.	Corrector beats raw CHIRPS (pinball -43% at q90; bias on extremes -66.6 → +4.8 mm at q99). Blind rule and max-of-sources are tested.	Wayanad gap not closed (71 vs 179 mm); IMD-vs-CHIRPS disagreement features did not improve cross-validation, so they are used only in the alert logic.
F2 Sensor faults undetected or confused with real extremes	Stage-1 engine with L1 rules, learned cross-source model, arbiter that retains corroborated extremes, and trust/repair outputs.	Macro recall 0.70 vs 0.30; zero-reading faults 0.80 vs 0.03-0.19; real-flag sanity 9/10; extremes retained.	Drift undetected; false-flag rate 4.9% on out-of-distribution stations (guard added in the alert engine, not in the model).
F3 Static thresholds: low detection, many false alarms	Learned model on rain history, wetness and terrain with monotone physics-consistent constraints; calibrated; recall-first operating point from out-of-fold data.	Cross-validated false alarms 18.8% vs 32.1-37.7% at 90% recall; unseen 2018: 21.4% vs 32.4-40.9%.	Target recall missed by the model alone; 80 test positives limit certainty.
F4 Inflated accuracy, biased labels	Frozen, checksummed, time-blocked splits; year-blocked cross-validation; single-look test; metadata never used as features; negatives matched and buffered; bootstrap intervals; baselines under the same protocol.	The first protocol's optimistic threshold was caught, disclosed and replaced; a post-evaluation sign bug was found, fixed and logged.	Inventory remains partial and uneven; false alarms are upper bounds.
F5 No uncertainty or degraded mode	10-90% interval from a 20-member ensemble, uncertain_high rule, data-confidence flag, out-of-distribution guard, reasons with every alert.	Interval rule adds 5 points of recall at Watch; confidence flags 23% of 2018 universe cell-days as Degraded (sources disagree, stations flagged).	Interval calibration is not claimed for events (too few).
Day-boundary mismatch between products	All rain converted to the IMD day with a tested approximation; day basis verified by lag correlation.	Station-vs-IMD peaks at lag 0; CHIRPS-UTC at -1, realigned CHIRPS at 0; approximation error 1.85 mm mean.	Hourly intensity is not used (daily resolution only).

8 Limitations and honest shortfalls

- **Not an operational system.** No claim of readiness; no live feed of landslide-site sensors exists openly for the region.
- **Statistical power.** The unseen test year has 80 in-domain positive cell-days (48 Himalaya, 32 Ghats); most comparisons between scorers cannot be resolved; per-region and leave-region-out results carry fewer than 100 positives.
- **Labels.** NASA catalogs are incomplete and rain-triggered only; unlabeled landslides sit among the negatives, so false-alarm shares are upper bounds. The recording rate varies more than 10-fold across years; post-2018 labels are nearly absent; 35 positives below the slope floor are outside the model's domain.
- **Calibration.** A pooled base rate cannot follow a time-varying inventory rate; probabilities are uncertain by about a factor of two and over-/under-predict by region (Himalaya over, Ghats about 3× under in 2018).
- **Recall target.** The hazard model alone reached 85.0% against a 90% target; the 91.2% figure for the full system depends on the uncertainty rule, which alerts on 27% of cell-days.
- **Wayanad.** The levels saturate and cannot single out the disaster day; CHIRPS reads 0 mm on that day (a product artifact); the corrector and ERA5 under-estimate.
- **Stage 1 is evaluated on injected, not real, faults** (only 10 real flags in 2007-2024); only 11 test and 11 validation stations carry episodes; slow drift is essentially undetected; the combined engine's false-flag rate is 4.9% on test because of two out-of-distribution Nepal gauges.
- **Reference weaknesses.** CHIRPS-realigned correlates only about 0.38 with gauges; IMD is not independent of the gauges; neighbour-gauge references are scarce (5.1% of clean days at the design radius).
- **Resolution.** Daily 0.05° data: no hourly intensity, so short intense bursts are not resolved; terrain is 90 m; lithology, soil and land cover are not used.
- **Elevation** is the top input and dropping it worsens false alarms; it may partly act as a location or reporting proxy (leave-region-out is indicative only).
- **Corrector** is trained on a few mostly lowland gauges and cannot be assumed to transfer to windward slopes.
- **Process disclosures.** The first model protocol's test result prompted the second protocol; one sign error in a displayed paired-difference value was fixed after the test evaluation (the model and all conclusions were unchanged, and the extra run is logged).

9 How the system can be improved

Ordered by expected value for the effort, using the evidence in this report.

#	Improvement	Why (evidence in this project)	Expected benefit	Effort
1	**Add sub-daily rain** (IMERG half-hourly, GSMaP, ERA5 hourly) and compute peak 1-6 hour intensity features.	The design called for hourly intensity but only daily data were feasible; rain_acc_1 is already the top rain feature; Wayanad rain fell in hours; the day-boundary issue is partly a resolution issue.	Higher detection of short, intense bursts; fewer false alarms from long drizzle; better lead-time logic.	Medium
2	**Denser, better gauge data:** IMD automatic weather/rain stations, state networks (e.g. Kerala, Uttarakhand), Nepal DHM stations; keep their real quality flags.	Only 140 GHCN gauges, 26.5% populated; the nearest trusted gauge was the best source at Wayanad (155 mm vs 179 mm IMD); neighbour references usable on only 5.1% of clean days.	Real fault labels for Stage 1; usable station-trust features; better extremes.	High (access, cleaning)
3	**Better and denser landslide labels:** post-event inventories (GSI, ISRO-NRSC landslide atlas, Sentinel-1/2 change-detection maps, state disaster reports), with event time of day; include 2019-2024.	80 test positives; inventory rate varies 10-fold by year; post-2018 labels near zero; Wayanad unlabelled.	Tighter intervals, trustworthy calibration, a real second test year, Wayanad scored.	High
4	**Fix calibration** with a time-varying or region-specific base rate and conformal intervals; report per-region reliability.	Ghats under-predicted ≈ 3×, Himalaya over-predicted; yearly observed rates vary > 10×.	Probabilities usable for cost-loss decisions.	Low-medium

#	Improvement	Why (evidence in this project)	Expected benefit	Effort
5	**Region-aware alert levels** (relative to local climatology or percentiles of risk) and regional models/thresholds.	Wayanad focus cell is at Emergency on every day; Ghats vs Himalaya behave differently (leave-region-out fails Ghats → Himalaya).	Levels that discriminate days within a wet spell.	Low-medium
6	**Orographic and atmospheric predictors:** windward exposure and upslope wind, moisture flux, IMD/ECMWF rain forecasts for the next 1-3 days.	Global products miss orographic extremes (F1); corrector cannot transfer to windward slopes; the system is a same-day nowcast only.	Better extreme-rain estimates and real lead time.	Medium-high
7	**Richer static and dynamic layers:** lithology (GSI), soil depth/type (SoilGrids), land cover/roads (WorldCover, OSM), satellite soil moisture (SMAP, ERA5-Land), snow/glacier melt for the Himalaya, rainfall windows of 60-90 days.	Only terrain and coarse MERRA-2 wetness are used; elevation acts as a catch-all.	Less reliance on elevation; better physical sense.	Medium
8	**Make Stage 1 robust to new stations:** train with more diverse (including Nepal-like, high-elevation) stations; per-station adaptive thresholds or conformal calibration; add change-point detectors (CUSUM) for slow drift; promote the out-of-distribution guard into the model.	91% of test false flags came from two Nepal gauges; drift recall ≈ 0.02-0.22.	False-flag rate at target on new stations; drift detected.	Medium
9	**Spatial block cross-validation and more regions** (Uttarakhand, Sikkim, north-east, Nepal, Sri Lanka) with model comparison against sequence models and physically based slope-stability (infinite-slope / TRIGRS) hybrids.	Leave-region-out shows poor transfer; only two regions; tree models only.	Know where the model generalises; hybrid physics gains.	High
10	**Operational layer:** real-time IMD/IMERG ingest, scheduled runs, monitoring of drift in inputs and alert rate, model registry, container image, authentication, shadow-mode trial with a disaster agency, and co-designed alert wording and cost-loss ratios.	The live mode is a labelled demo using unvalidated inputs; the cost-loss analysis shows the levels depend on how cheap acting is.	Path from prototype to a pilot.	High

Quick wins that need no new data: items 4 and 5 (calibration and region-aware levels), the out-of-distribution guard moved into Stage 1 (item 8), and a prospective-style replay for 2019-2024 using whatever labels can be assembled. **Highest-value investments** are items 1-3: better rain resolution, more gauges and more labels attack the three weaknesses that limit every number in this report.

10 Reproducing and running the system

Everything lives in `C:\Users\sujan\AtherGuard`. The pipeline runs stage by stage from `configs/pipeline.yaml`; stages whose frozen outputs verify intact are reused, so re-runs are fast.

```
cd C:\Users\sujan\AtherGuard
.venv\Scripts\python -m aetherguard.pipeline status
.venv\Scripts\python -m aetherguard.pipeline run --dry-run
.venv\Scripts\python -m aetherguard.pipeline run [--from STAGE] [--to STAGE] [--only STAGE] [--offline]
[--force]
.venv\Scripts\python -m aetherguard.finalreport
```

Serving:

```
.venv\Scripts\python -m aetherguard.serve.api --port 8765
.venv\Scripts\streamlit run src\aetherguard\serve\dashboard.py --server.address 127.0.0.1
```

Tests: `.venv\Scripts\python -m pytest -q` runs 1,323 offline tests; real-data and live tests are opt-in via the `realdata` and `live` markers. Re-downloading the NASA label catalogs needs a free Earthdata login (stored in a netrc file, never in the project).

Stage	Module	Network
labels, splits	events.build, splits.build (+verify)	no
rain, wetness, terrain	ingest.imd_build, chirps_build, power_build, terrain_build; features.regions_build	yes
samples, features	features.samples_build, features_build (+verify)	no
Stage 1	integrity.baseline_report, engine_data, day_basis, engine_report, hazard_features	no
corrector	corrector.report	no
hazard model	hazard.train, hazard.freeze, hazard.evaluate_report (single frozen evaluation)	no
cases, alerts	cases.series; alerts.derive, alerts.replay, alerts.report	cases: yes
final report	finalreport	no

Appendix A Glossary

Term	Meaning
Cell-day	One 0.05° (~5 km) cell on one calendar day: the unit the model scores.
Positive / negative	A cell-day with (positive) or without (negative) a recorded rain-triggered landslide.
Universe	A weighted random sample of all cell-days, used to measure false-alarm rates and set thresholds.
Recall / detection rate	Share of real events that were flagged.
False-alarm share	Share of all cell-days on alert (an upper bound here because labels are incomplete).
Monotone constraint	A rule forcing the model to never decrease risk when rain (or slope) increases.
Isotonic calibration	A monotone mapping that turns model scores into probabilities that match observed frequencies.
Out-of-fold	A prediction made by a model that did not see that row's year or station.
Year-blocked cross-validation	Training on some years and scoring the held-out years, so neighbouring days never leak across folds.
Pinball loss	The loss for quantile prediction; lower means the predicted quantile is better placed.
IMD day	08:30 IST to 08:30 IST; the day basis all rain is converted to.
LHASA	NASA's Landslide Hazard Assessment for Situational Awareness (susceptibility × rain).
SHAP	Per-prediction contributions of each input, used for the 'why this alert' view.
Frozen	Fixed and checksummed so it cannot change after results are seen.
GHCN-D, IMD, CHIRPS, MERRA-2, ERA5, GLO-90	Gauge archive, India's gridded rain, satellite rain, NASA reanalysis, ECMWF reanalysis, Copernicus 90 m terrain.

Appendix B How the project was built

Phase	What was done	Key outcome
0 Data audit	Probed every source (access, size, licence, coverage); built the environment, manifest and audit report	Showed which assumptions fail (labels end in 2018, 2 Nepal gauges, thin flags) and redesigned the test sets
1 Ingest and features	Label store, frozen splits, IMD / CHIRPS (14.97 GB) / MERRA-2 / terrain ingest, regions, negatives, universe sample, feature store	919 positives, 166,728 cells, frozen read-only data with checksums
2 Sensor integrity	Fault injection into real series, 4 baselines, neighbour/grid references, day-basis experiment, LightGBM engine, arbiter, repair, out-of-fold outputs	Macro recall 0.70 vs 0.30; Nepal out-of-distribution shortfall reported
3 Hazard, corrector, cases	Hazard model v1 → protocol v2, corrector, ERA5 case inputs, case scorer	85.0% / 21.4% on unseen 2018; target missed; Wayanad gap not closed
4 Alerts, API, dashboard	Alert engine, replay store, FastAPI, Streamlit, alerts report	91.2% at Watch+; Wayanad saturation reported
5 Final report and runner	24-stage pipeline runner, final-report generator, README	One-command reproduction; every number read from a file

Process and disclosures. Work followed a plan-build-review loop: an architect model decided designs and reviewed results, builder models implemented and tested, and every module was tested first and verified against real data. Three disclosures matter for trust: (1) **Protocol change.** The first hazard-model protocol re-used one validation split for selection, calibration and the threshold; its test result (78.7% recall; ties with baselines) prompted protocol v2 with cross-validated selection, and the report says so. (2) **Controller rulings after seeing results** were limited to reporting and safety rules (no retraining after seeing test): the Stage-1 shortfall was reported, not tuned away; station-presence columns were excluded from model inputs before group selection; the Wayanad saturation was recorded as a limitation. (3) **A bug fix after the test run:** a sign error in the point value of one paired recall difference was found in review, fixed with a regression test, and the evaluation re-run once with the model frozen and unchanged (logged in the freeze record).

Appendix C Key files

File or folder	What it holds
reports/FINAL_REPORT.md / .json	Assembled results, every figure traced to a source file (535 figures)
reports/phase0_data_audit.md, phase0_findings.md	Data audit and findings
reports/phase2_integrity_report.md, phase2a_baselines.md, phase2b_day_basis.md	Stage 1 results, baselines, day-basis experiment
reports/phase3_hazard_report.md, phase3_hazard_cv_report.md, phase3_hazard_report_v1.md	Frozen test evaluation, cross-validation, first-protocol history
reports/phase3_corrector_report.md, phase3e_case_inputs.md, phase4_alerts_report.md	Corrector, case inputs, alert-system results
docs/phase1-design.md … phase4-design.md, docs/superpowers/specs/…-design.md	Design decisions, rulings and findings per phase; the original spec
data/processed/hazard/model/ (card.json, FROZEN.json, TEST_EVALUATION.json)	Frozen hazard model, thresholds, freeze checksums, evaluation log
data/processed/features, splits, labels, integrity, alerts, cases	Frozen features and splits, label store, engine outputs, replay store, case inputs
configs/*.yaml; src/aetherguard/…; tests/…	Configuration, source code, 1,323 tests